

Infrastructure Projects

Infrastructure Projects is where you provision real cloud infrastructure using Terraform and Ansible. Compose your infrastructure from modular services — VPC, security groups, key pairs, EC2 instances and more — then execute directly against your cloud account and watch the live output stream in real time.

What are Infrastructure Projects?

An Infrastructure Project is a collection of configured cloud services that together define a complete environment. Each service maps to a Terraform module — for example a VPC module, a security group module, a key pair module, or an EC2 instance module. You compose these services inside a project, configure them to your requirements, and then execute the project against your cloud account to provision the actual infrastructure.

QuickInfra handles the Terraform execution engine underneath — you do not need Terraform installed locally or any CLI access. Select your cloud account, choose a Terraform action, hit Execute, and the output streams live to your terminal panel inside the platform. Ansible playbooks can also be layered on top for post-provisioning configuration management.

 **Note:** Infrastructure Projects use **OpenTofu** — the open-source fork of Terraform — as the execution engine. All standard Terraform module syntax and providers are fully supported.

Three Ways to Create a Project

From Scratch — Create a blank project, give it a name and region, then manually add and configure each service module. Best when you want full control over every parameter.

Copy from an Existing Project — Clone any existing project — including production — as a starting point. All services and configurations are duplicated instantly. Change what you need and you have a ready-to-deploy replica. This is the fastest way to replicate production infrastructure into a staging or dev account.

From a Ready-Made Template — QuickInfra provides a library of pre-configured infrastructure templates covering common patterns. Select one and a fully configured project with all services already set up is created for you — ready to execute with minimal changes.



Tip: A project is not tied to a single cloud account. Configure your infrastructure once, execute it in your dev account to test, then switch the cloud account to staging or production and execute again — identical infrastructure deployed across any number of environments with zero rework.

Project Details - EC2 DevOps Infra Simple (Mumbai)

Home > Projects > Infrastructure > Project Details > Service

Configured Services (5) MFA (Optional) + Add Service

Service Type	Module Name	Description	Status	Actions
VPC	SimpleVPC	Simple VPC Network	Active	
SG	SG-Web	SG-Web	Active	
KeyPair-EC2	SingleAZ_Key	SingleAZ_Key	Active	
EC2	EC2-Blue	EC2-Blue	Active	
EC2	EC2-Green	EC2-Green	Active	

Rows per page: 10 1-5 of 5

Select Cloud Account: AWS Account Dev Select Terraform Action: Select Action Execute Connected

Tip: Click the Terminal icon in the navbar to open the Terraform console and view execution output.

A configured Infrastructure Project — five services (VPC, Security Group, KeyPair, EC2-Blue, EC2-Green) compose a complete DevOps infrastructure. Select a cloud account and Terraform action, then execute.

How to Set Up and Execute a Project

1 Create an Infrastructure Project

Navigate to **Projects** → **Infrastructure** and click **New Project**. Choose to start from scratch, copy an existing project, or pick a ready-made template. Give the project a name that reflects the environment — for example *EC2 DevOps Infra Simple (Mumbai)* — and select the target cloud provider and region.

2 Add services to the project

Inside the project, click **Add Service** to add the first infrastructure module. Each service has a **Service Type** — such as VPC, SG, KeyPair-EC2, or EC2 — and a **Module Name** that identifies this specific instance of that type. Configure the module parameters for each service added.

3 Build your service stack

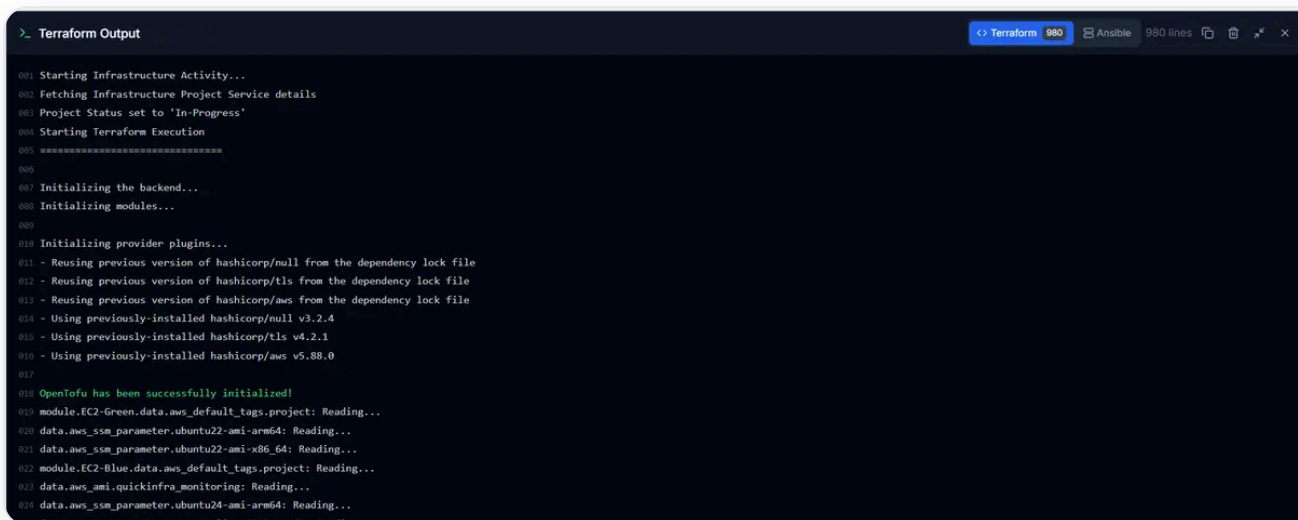
Add all the services your environment requires. A typical stack follows a dependency order — VPC first, then security groups, then key pairs, then compute instances. QuickInfra manages the dependency graph during execution so resources are created in the correct sequence.

4 Select cloud account and action

At the bottom of the project services screen, select the **Cloud Account** to deploy into and choose a **Terraform Action** — **Plan** to preview changes without applying, **Apply** to provision resources, or **Destroy** to tear down all resources in the project.

5 Execute and monitor live output

Click **Execute**. QuickInfra streams the Terraform execution output live to the terminal panel. You can switch between the **Terraform** and **Ansible** output tabs to monitor each layer of the execution independently.



```
> Terraform Output
[Terraform 980] [Ansible 980 lines]

001 Starting Infrastructure Activity...
002 Fetching Infrastructure Project Service details
003 Project Status set to 'In-Progress'
004 Starting Terraform Execution
005 =====
006
007 Initializing the backend...
008 Initializing modules...
009
010 Initializing provider plugins...
011 - Reusing previous version of hashicorp/null from the dependency lock file
012 - Reusing previous version of hashicorp/tls from the dependency lock file
013 - Reusing previous version of hashicorp/aws from the dependency lock file
014 - Using previously-installed hashicorp/null v3.2.4
015 - Using previously-installed hashicorp/tls v4.2.1
016 - Using previously-installed hashicorp/aws v5.88.0
017
018 OpenTofu has been successfully initialized!
019 module.EC2-Green.data.aws_default_tags.project: Reading...
020 data.aws_ssm_parameter.ubuntu22-ami-arm64: Reading...
021 data.aws_ssm_parameter.ubuntu22-ami-x86_64: Reading...
022 module.EC2-Blue.data.aws_default_tags.project: Reading...
023 data.aws_ami.quickinfra_monitoring: Reading...
024 data.aws_ssm_parameter.ubuntu24-ami-arm64: Reading...
025 data.aws_ssm_parameter.ubuntu24-ami-x86_64: Reading...
```

Live Terraform execution output — QuickInfra streams every line of the OpenTofu run in real time, with line numbers and colour-coded status messages.

Terraform Actions

The **Select Action** dropdown gives you full control over what happens when you hit Execute. Each action maps to a specific Terraform operation run against your cloud account.

Generate Terraform — Generates the Terraform configuration files for your project based on the services you have configured. No infrastructure is created yet — this just prepares the scripts and validates your configuration.

View Terraform Plan — Shows a preview of what will be created, modified, or destroyed when you execute against your cloud account. Nothing changes in AWS — this is a safe read-only check and is recommended before every Apply in production.

Create Infrastructure — Applies your Terraform configuration and provisions the actual cloud resources in your account. VPCs, EC2 instances, security groups, and all configured services get created.

Plan and Create — Runs a plan first, then immediately applies it in one step. Recommended for most use cases — you get the safety of a plan with the convenience of a single action.

Remove Infrastructure — Destroys all cloud resources associated with this project in your account. This permanently deletes VPCs, EC2 instances, security groups, and all other provisioned resources. Use with caution in production.

Infrastructure Pricing — Estimates the monthly cost of your infrastructure before creating it. Sends your plan to the cost estimation engine and returns a breakdown per resource so you can review costs before committing.

Infrastructure Diff — Compares your current state file against the actual state of resources in your cloud account. Useful for detecting drift — changes made directly in AWS outside of QuickInfra will be surfaced here.

Update Modules — Refreshes the Terraform modules referenced in your project to their latest versions. Run this if a module has been updated and you want to pull in the latest changes before your next execution.

Upgrade Terraform — Upgrades the OpenTofu providers to their latest compatible versions as defined in your configuration. Run this to stay current with AWS provider updates.

✔ **Tip:** Run **View Terraform Plan** before every **Create Infrastructure** in production environments. The plan output shows exactly which resources will be created, modified, or destroyed — giving you a chance to catch unintended changes before they affect live infrastructure.

Service Actions

Each service in the project list has three action buttons available on the right:

Clone — Duplicate the service configuration as a starting point for a similar module. Useful when you need two identical EC2 modules with minor differences, such as EC2-Blue and EC2-Green.

View / Edit — Open the service configuration to review or modify its Terraform module parameters before the next execution.

Delete — Remove the service from the project. This does not destroy the cloud resource — use Terraform Destroy for that.

⚠ **Warning:** The **Remove Infrastructure** action will permanently delete all cloud resources associated with the project in your cloud account. This includes VPCs, EC2 instances, security groups, and all other provisioned resources. Use with caution in production.

Key Things to Know

The **Connected** indicator at the bottom right confirms QuickInfra has a live connection to your cloud account and is ready to execute.

Terraform state is managed by QuickInfra — you do not need to configure a remote backend or manage state files manually.

The terminal panel is accessible at any time via the terminal icon in the navbar, allowing you to review execution output even after navigating away from the project.

Ansible playbooks run after Terraform completes — they are used for post-provisioning configuration such as installing software, setting up users, and applying system hardening.

Each project is scoped to a single region — create separate projects for infrastructure in different regions or accounts.